

Nifty 100 Analytics Dashboard

Analyst User Guide

**PREPARED BY
MOHAMMAD ESA MOHIBUDDIN**

**BLUESTOCK FINTECH INTERNSHIP – CAPSTONE PROJECT
VERSION 1.0
JUNE 2026**

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About This Guide

This guide explains how to use the **Nifty 100 Analytics Dashboard**. It provides an overview of the dashboard features, navigation, and REST API endpoints to help users analyze financial data efficiently.

The guide is intended for financial analysts, researchers, and anyone using the dashboard for company analysis, peer comparison, stock screening, and sector analysis.

Notes

- Dashboard screenshots are included where applicable.
- API examples use the local development server.
- Financial data is retrieved from the project's SQLite database.

Project Overview

The **Nifty 100 Analytics Dashboard** is a financial analytics platform developed as part of the **Bluestock Fintech Internship Capstone Project**. The application provides an interactive interface for analyzing financial data of companies listed in the **Nifty 100 Index**.

The platform combines financial statements, key performance indicators (KPIs), valuation metrics, peer comparisons, sector analysis, and portfolio insights into a single dashboard. It enables users to access and visualize financial information efficiently without manually processing large datasets.

Project Objectives

The primary objectives of this project are:

- Develop a centralized dashboard for analyzing Nifty 100 companies.
- Provide quick access to financial statements and key ratios.
- Enable comparison of companies with their industry peers.
- Support stock screening using financial metrics.
- Present sector-level insights through interactive visualizations.
- Expose financial data through REST APIs for integration with other applications.

Key Features

- Interactive dashboard built with **Streamlit**
- REST API developed using **FastAPI**
- Financial data stored in **SQLite**
- Company profile and financial statement analysis
- Peer comparison dashboard
- Stock screener with financial filters
- Sector analysis and capital allocation visualizations
- Annual report access and company document links
- Interactive charts using **Plotly**

Target Users

This dashboard is designed for:

- Financial Analysts
- Equity Research Analysts
- Investors
- Students learning financial analysis
- Anyone interested in exploring the financial performance of Nifty 100 companies

Technology Stack

Component	Technology
Programming Language	Python
Dashboard	Streamlit
Backend API	FastAPI
Database	SQLite
Data Processing	Pandas
Visualizations	Plotly
Testing	Pytest
Code Quality	Black, Ruff

Home Page

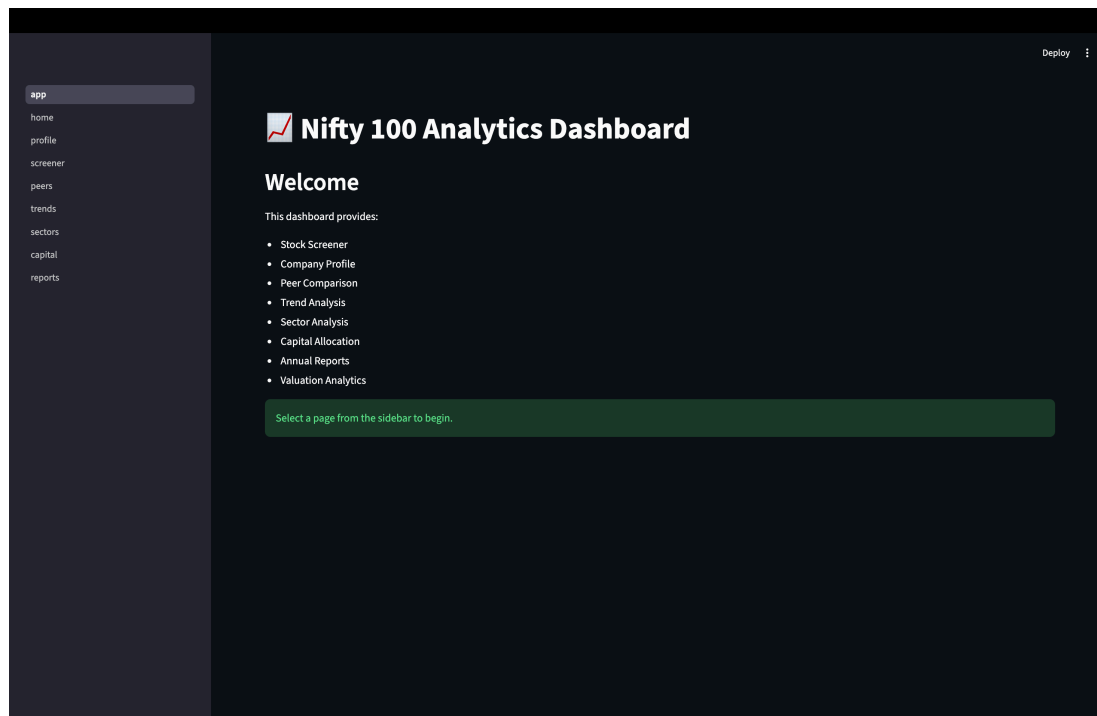


Figure 1: Home Dashboard displaying key financial insights and navigation modules of the Nifty 100 Analytics Dashboard.

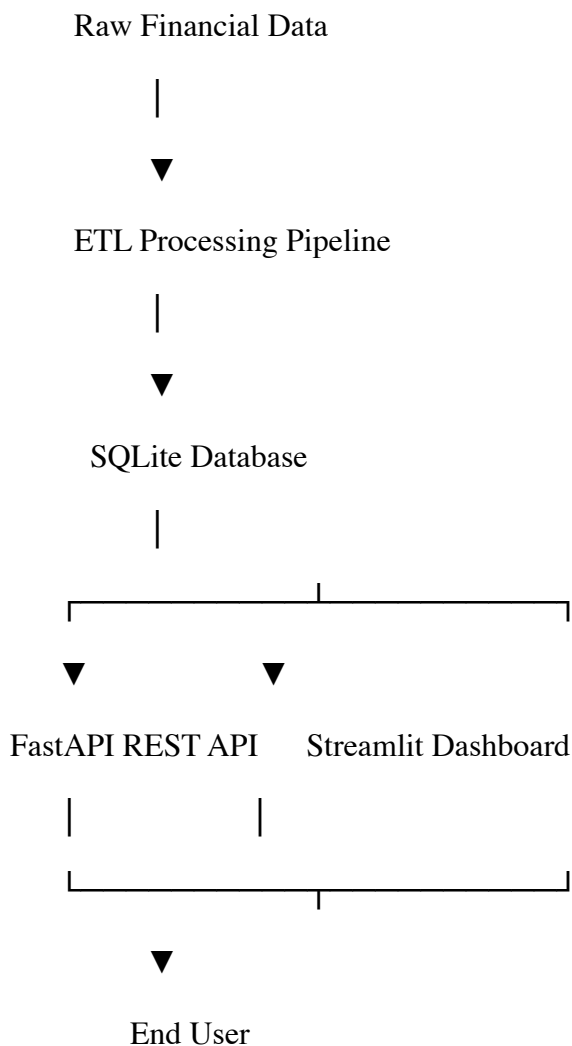
System Architecture

The **Nifty 100 Analytics Dashboard** follows a modular architecture that separates data processing, storage, backend services, and user interface components. This design improves maintainability, scalability, and performance.

The application processes raw financial data through an ETL pipeline, stores the processed data in a SQLite database, exposes the data through REST APIs built with FastAPI, and presents interactive visualizations using Streamlit.

Architecture Diagram

Create the following simple flow diagram in Apple Pages using **Shapes** (Rectangles + Arrows):



Components

ETL Pipeline

The ETL (Extract, Transform, Load) pipeline imports raw financial datasets, validates the data, computes financial ratios, and stores the processed information in the SQLite database.

SQLite Database

SQLite serves as the central data repository for the application. It stores company profiles, financial statements, ratios, sector information, valuation data, peer groups, and other analytical datasets.

FastAPI

FastAPI provides REST API endpoints that allow the dashboard and external applications to retrieve financial data efficiently. It offers fast performance and automatic API documentation.

Streamlit Dashboard

The Streamlit application provides an interactive user interface where users can explore companies, compare financial performance, analyze sectors, screen stocks, and visualize trends through charts and tables.

End User

Financial analysts, researchers, investors, and students interact with the dashboard to analyze financial information and gain insights from the available data.

Data Flow

The workflow of the application is as follows:

1. Financial data is collected from source datasets.
2. The ETL pipeline validates and transforms the data.

3. Processed data is stored in the SQLite database.
4. FastAPI exposes the data through REST API endpoints.
5. Streamlit retrieves the data and displays interactive dashboards.
6. Users analyze companies through visualizations and reports.

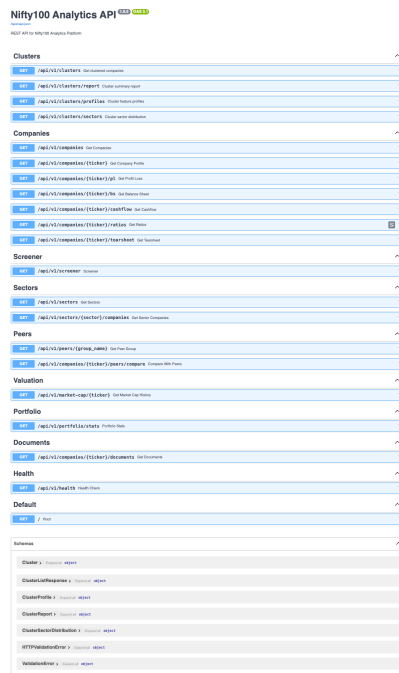


Figure 2: FastAPI Swagger documentation displaying the available REST API endpoints for the Nifty 100 Analytics Dashboard.

Dashboard Navigation

The **Nifty 100 Analytics Dashboard** provides a user-friendly interface that allows users to access various financial analysis modules from a single platform. Each module focuses on a specific aspect of company or market analysis, enabling users to explore financial data efficiently.

The navigation menu allows users to switch between different analytical views without leaving the application.

Dashboard Modules

Module	Description
Home	Displays an overview of the dashboard and key financial insights.
Company Profile	View company information, financial statements, and key performance indicators.
Stock Screener	Filter companies using financial ratios and performance metrics.
Peer Comparison	Compare a company with its industry peers using financial metrics.
Trend Analysis	Analyze historical trends through interactive charts and graphs.
Sector Analysis	Compare companies across different sectors and industries.
Capital Allocation	Visualize how companies allocate capital across various financial activities.
Annual Reports	Access company reports and supporting financial documents.

Navigation Workflow

Users can navigate through the dashboard using the sidebar menu. After selecting a module, the dashboard displays the corresponding data, charts, and analysis tools. Filters and interactive visualizations allow users to explore information dynamically without reloading the application.

Key Features

- Simple and intuitive user interface
- Interactive charts and visualizations
- Company search functionality
- Financial ratio analysis
- Peer benchmarking
- Sector-wise insights
- REST API integration
- Responsive dashboard layout

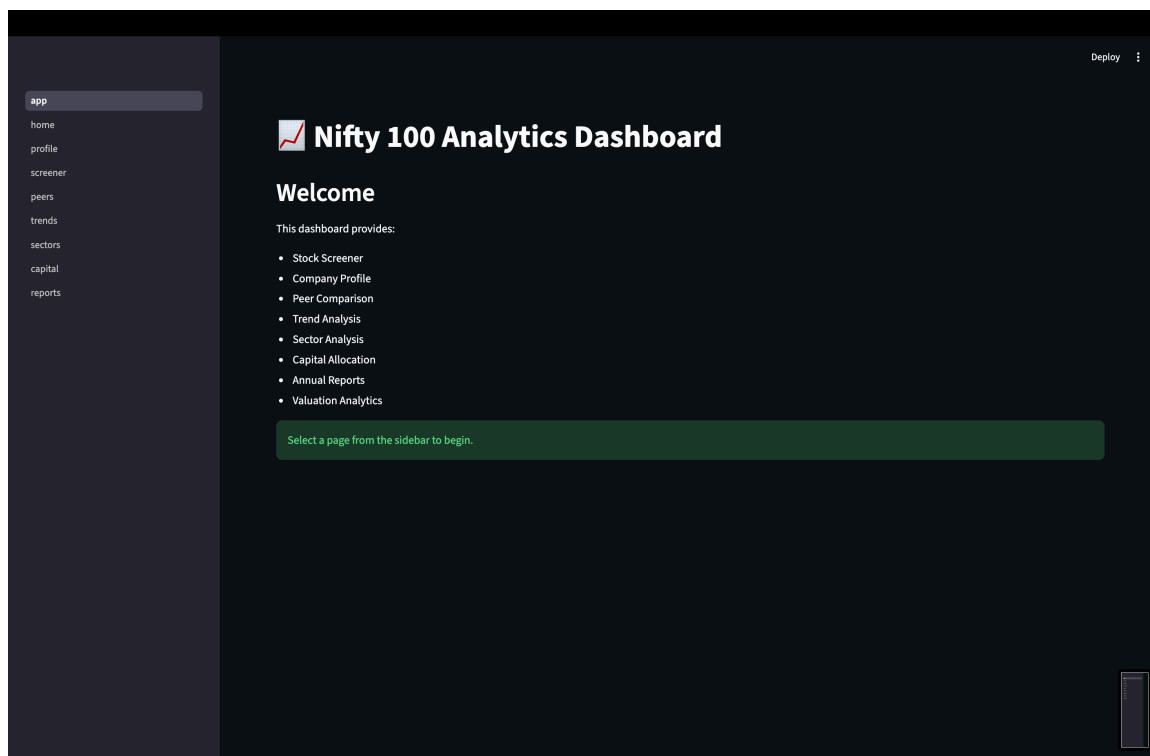


Figure 3: Dashboard navigation menu providing access to all major financial analysis modules.

Home Dashboard

The **Home Dashboard** serves as the main landing page of the Nifty 100 Analytics Dashboard. It provides users with a high-level overview of the platform and quick access to the different financial analysis modules.

The home page is designed to help users begin their analysis by presenting key information and navigation options in a simple and organized layout.

Purpose

The Home Dashboard enables users to:

- Access all dashboard modules from a single location.
- View an overview of the financial analytics platform.
- Navigate quickly to company analysis and other features.
- Start financial research with an intuitive interface.

Features

The Home Dashboard includes the following features:

- Dashboard title and introduction
- Navigation menu for all analysis modules
- Quick access to financial analytics tools
- User-friendly interface with interactive components
- Consistent layout for seamless navigation

How to Use

1. Launch the Streamlit application.
2. Open the Home Dashboard.
3. Use the sidebar navigation menu to select a module.
4. Explore the available financial analysis features.
5. Continue to other modules such as Company Profile, Stock Screener, or Peer Comparison.

Benefits

The Home Dashboard offers several advantages:

- Easy navigation across the application.

- Centralized access to all dashboard modules.
- Clean and intuitive interface.
- Improved user experience for financial analysis.

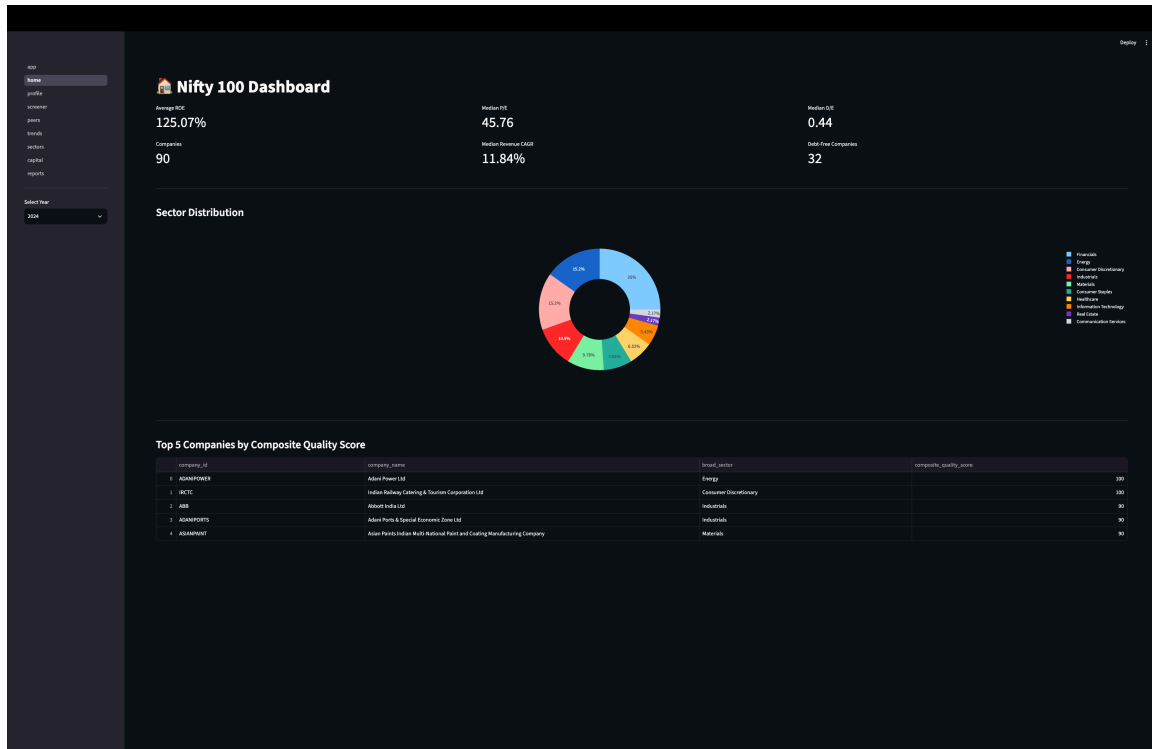


Figure 4: Home Dashboard displaying the main interface and navigation options of the Nifty 100 Analytics Dashboard.

Company Profile

The **Company Profile** module provides a comprehensive view of an individual company's financial information. It brings together company details, financial statements, key financial ratios, and performance metrics in a single interface, enabling users to perform detailed financial analysis.

Users can select a company from the available list and explore its financial performance through organized tables and interactive visualizations.

Purpose

The Company Profile module helps users to:

- Analyze the financial performance of a selected company.
- Review company fundamentals and financial statements.
- Compare financial metrics across multiple years.
- Understand profitability, liquidity, leverage, and operational efficiency.
- Support investment research and decision-making.

Features

The Company Profile module includes the following information:

- Company details and basic information
- Profit & Loss Statement
- Balance Sheet
- Cash Flow Statement
- Financial Ratios
- Key Performance Indicators (KPIs)
- Interactive charts and financial trends
- Company documents and annual report links (if available)

How to Use

1. Open the **Company Profile** module from the navigation menu.
2. Select a company from the available list.
3. Review the company details and financial statements.
4. Analyze financial ratios and key performance indicators.
5. Use the charts to understand historical financial trends.

Benefits

The Company Profile module allows users to:

- Evaluate a company's financial health.
- Identify long-term financial trends.
- Review profitability and operational performance.
- Access important financial information from a single dashboard.



Figure 5: Company Profile module displaying company information, financial statements, key performance indicators, and financial ratios.

Stock Screener

The **Stock Screener** module enables users to filter and identify companies based on various financial metrics and performance indicators. It helps analysts quickly narrow down companies that meet specific investment or research criteria without manually reviewing every company in the database.

This feature is useful for comparing multiple companies and identifying those with strong financial fundamentals.

Purpose

The Stock Screener module allows users to:

- Filter companies using financial ratios and performance metrics.
- Identify companies that match specific investment criteria.
- Compare multiple companies efficiently.
- Simplify the process of financial research.

Features

The Stock Screener provides:

- Financial ratio-based filtering
- Company search functionality
- Interactive results table
- Sorting and comparison of companies
- Quick access to company profiles
- Easy exploration of screened results

How to Use

1. Open the **Stock Screener** module from the navigation menu.
2. Select the desired financial filters (such as profitability or valuation metrics).
3. Apply the filters to generate a list of matching companies.
4. Review the results in the interactive table.
5. Select a company for further analysis in the Company Profile module.

Benefits

Using the Stock Screener helps users:

- Save time when analyzing multiple companies.
- Focus on companies that meet predefined financial conditions.
- Compare businesses using consistent financial metrics.
- Support investment research with data-driven filtering.

Common Use Cases

The Stock Screener can be used to:

- Find financially strong companies.
- Identify companies with healthy profitability.
- Compare valuation metrics across multiple companies.
- Create a shortlist for further financial analysis.

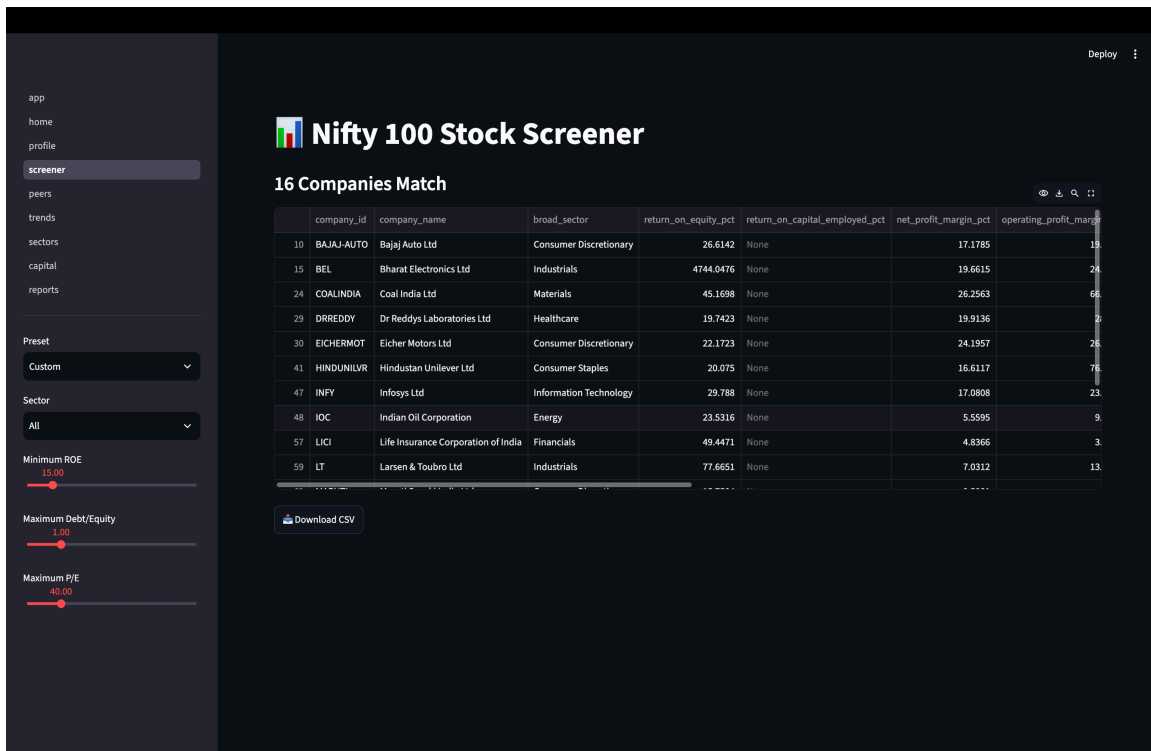


Figure 6: Stock Screener module displaying financial filters and the list of companies matching the selected criteria.

Peer Comparison

The **Peer Comparison** module enables users to compare a selected company with other companies in the same industry or peer group. By analyzing financial metrics side by side, users can evaluate relative performance, identify strengths and weaknesses, and gain a better understanding of a company's competitive position.

This module simplifies benchmarking by presenting key financial indicators in a structured and easy-to-read format.

Purpose

The Peer Comparison module helps users to:

- Compare a company with its industry peers.
- Evaluate financial performance using standardized metrics.
- Identify companies with stronger or weaker financial fundamentals.
- Support investment research and competitive analysis.

Features

The Peer Comparison module includes:

- Company selection
- Peer group identification
- Side-by-side financial comparison
- Key financial ratios and performance metrics
- Interactive tables and visualizations
- Comparative analysis for informed decision-making

How to Use

1. Open the **Peer Comparison** module from the navigation menu.
2. Select the company you want to analyze.
3. View the automatically generated peer group.
4. Compare financial metrics such as profitability, valuation, leverage, and efficiency.
5. Use the comparison to assess the company's position within its industry.

Benefits

The Peer Comparison module enables users to:

- Benchmark companies against competitors.
- Identify industry leaders based on financial performance.
- Understand competitive strengths and weaknesses.
- Make more informed financial and investment decisions.

Key Comparison Metrics

Depending on the available data, users can compare metrics such as:

- Revenue
- Net Profit
- Earnings Per Share (EPS)
- Return on Equity (ROE)
- Return on Capital Employed (ROCE)
- Price-to-Earnings (P/E) Ratio
- Debt-to-Equity Ratio
- Market Capitalization

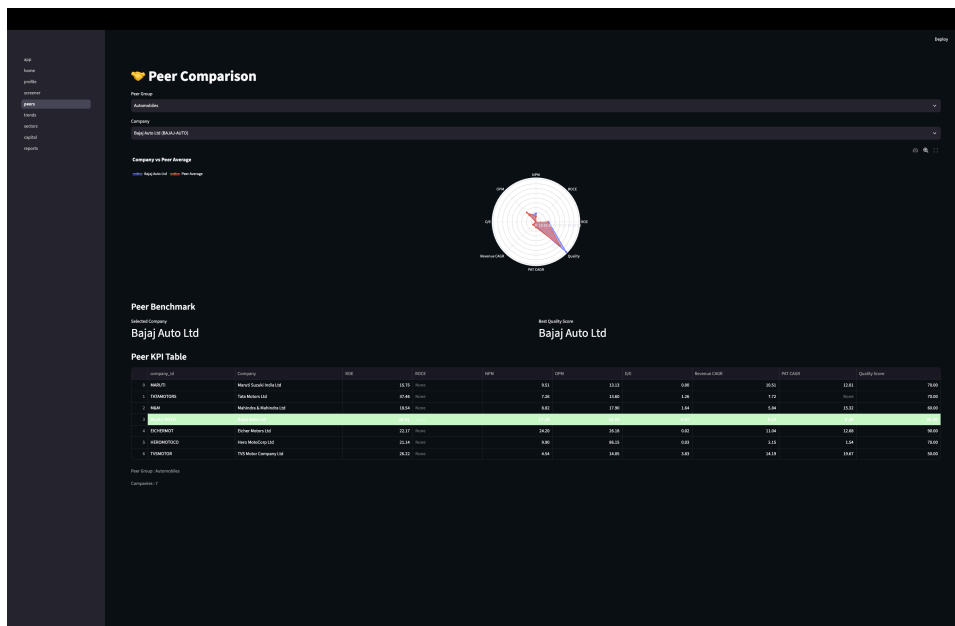


Figure 7: Peer Comparison module displaying financial metrics for a selected company and its industry peers.

Trend Analysis

The **Trend Analysis** module helps users evaluate the historical performance of a company by presenting financial data across multiple years. It enables users to identify growth patterns, performance changes, and long-term trends using interactive charts and tables.

By analyzing trends instead of a single year's data, users can make more informed financial assessments and better understand a company's overall performance.

Purpose

The Trend Analysis module allows users to:

- Analyze historical financial performance.
- Identify long-term growth or decline.
- Compare financial metrics across multiple years.
- Support data-driven investment and research decisions.

Features

The Trend Analysis module includes:

- Multi-year financial data
- Interactive line and bar charts
- Historical performance comparison
- Financial trend visualization
- Easy-to-read tables and graphs

How to Use

1. Open the **Trend Analysis** module from the navigation menu.
2. Select the company to analyze.
3. Review the available historical financial data.
4. Use the interactive charts to observe performance trends.
5. Compare different financial metrics over multiple years.

Benefits

The Trend Analysis module helps users to:

- Understand long-term financial performance.
- Identify consistent growth or declining trends.
- Detect changes in profitability and financial health.
- Support better investment and business decisions.

Common Metrics Analyzed

Users can review trends for metrics such as:

- Revenue
- Net Profit
- Earnings Per Share (EPS)
- Operating Profit
- Return on Equity (ROE)
- Return on Capital Employed (ROCE)
- Debt-to-Equity Ratio
- Cash Flow



Figure 8: Trend Analysis module displaying historical financial performance and interactive charts for a selected company.

Sector Analysis

Sector Analysis

The **Sector Analysis** module enables users to evaluate the financial performance of companies within different sectors of the Nifty 100 index. It provides insights into sector-wise trends, allowing users to compare industries and identify sectors with strong financial performance.

By grouping companies based on their sectors, this module helps users understand industry trends and supports informed investment and research decisions.

Purpose

The Sector Analysis module helps users to:

- Compare the performance of different sectors.
- Analyze sector-wise financial metrics.
- Identify high-performing and underperforming industries.
- Understand the distribution of companies across sectors.
- Support market and industry research.

Features

The Sector Analysis module includes:

- Sector-wise company distribution
- Financial performance comparison
- Interactive charts and visualizations
- Summary statistics for each sector
- Comparative analysis across industries

How to Use

1. Open the **Sector Analysis** module from the navigation menu.
2. Select a sector or review all available sectors.
3. Explore the charts and summary statistics.
4. Compare financial performance across different industries.
5. Use the insights to identify sectors for further analysis.

Benefits

The Sector Analysis module enables users to:

- Understand industry-level financial performance.
- Compare companies within the same sector.
- Identify sectors showing strong growth or stability.
- Support investment strategies based on sector trends.

Typical Insights

Users can gain insights such as:

- Number of companies in each sector.
- Overall financial performance by sector.
- Comparison of profitability and valuation metrics.
- Sector-wise trends and patterns.

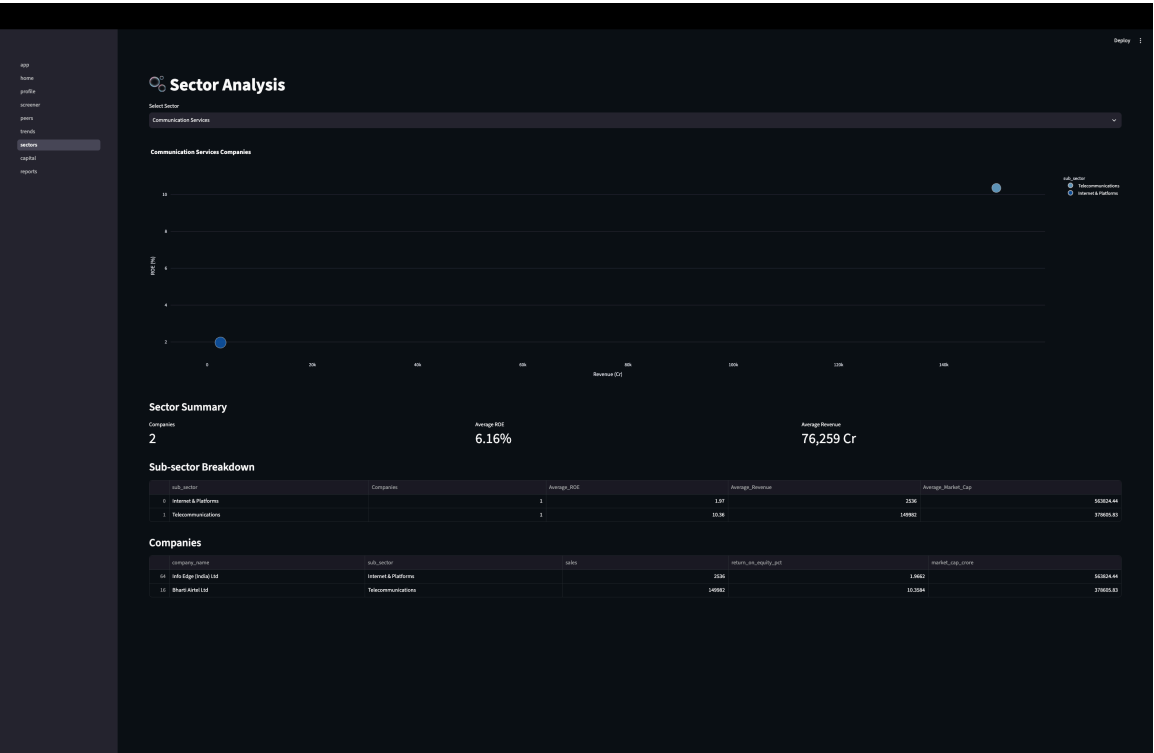


Figure 9: Sector Analysis module displaying sector-wise financial insights and comparative visualizations.

Capital Allocation

The **Capital Allocation** module provides insights into how companies utilize their financial resources. It helps users understand how capital is distributed across different areas such as business operations, investments, debt repayment, and shareholder returns.

By visualizing capital allocation patterns, users can evaluate a company's financial strategy and assess how effectively it manages its available resources.

Purpose

The Capital Allocation module helps users to:

- Analyze how companies allocate their capital.
- Understand investment and financing decisions.
- Evaluate the balance between business growth and shareholder returns.
- Support long-term financial analysis.

Features

The Capital Allocation module includes:

- Interactive charts and visualizations
- Company-wise capital allocation analysis
- Comparison of allocation patterns across companies
- Summary tables for financial insights
- Easy-to-understand graphical representation of financial data

How to Use

1. Open the **Capital Allocation** module from the navigation menu.
2. Select the company you wish to analyze.
3. Review the available charts and summary information.
4. Compare the company's capital allocation with other companies, if applicable.
5. Use the insights to better understand the company's financial strategy.

Benefits

The Capital Allocation module enables users to:

- Understand how companies prioritize the use of their financial resources.
- Identify companies investing in future growth.
- Evaluate financial management strategies.
- Support investment research with visual financial insights.

Typical Insights

Users can identify:

- Capital investment trends.
- Allocation towards business growth and expansion.
- Debt management and financing patterns.
- Overall capital utilization strategies.

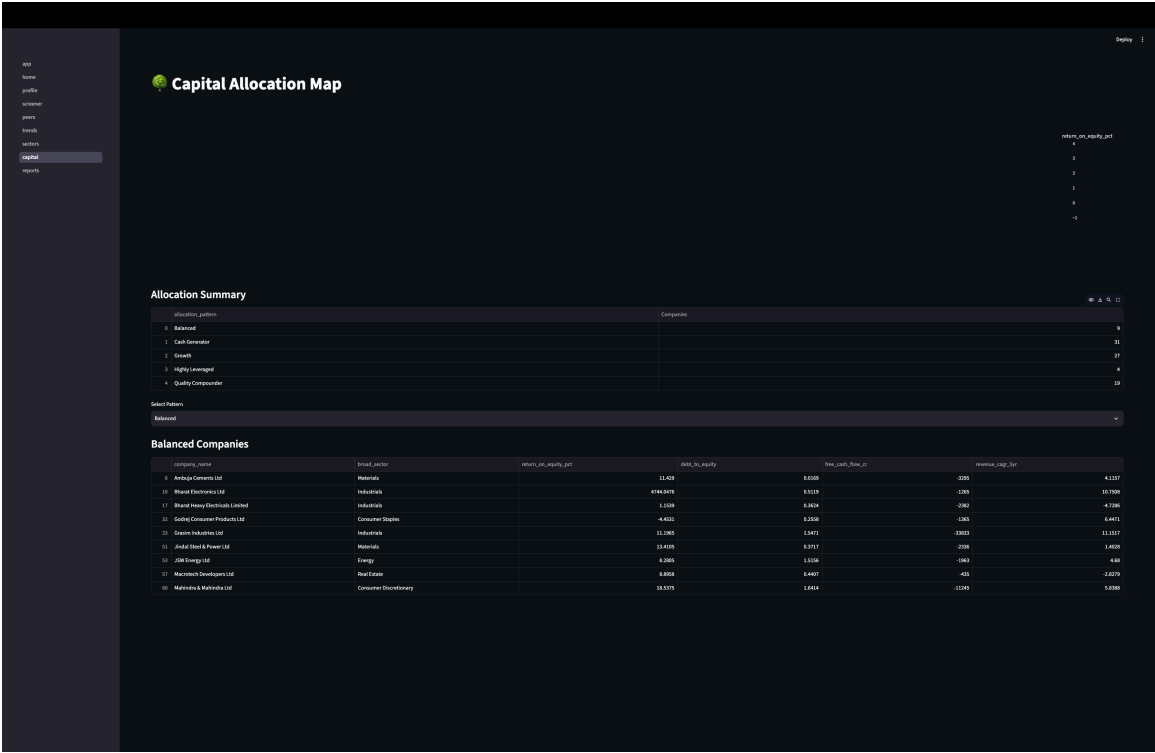


Figure 10: Capital Allocation module displaying visual insights into how companies allocate their financial resources.

Annual Reports

The **Annual Reports** module provides users with access to company reports and financial documents available within the dashboard. These reports contain detailed information about a company's financial performance, business operations, management discussion, corporate governance, and future outlook.

This module serves as a central location for accessing supporting documents that complement the financial analysis presented in other dashboard modules.

Purpose

The Annual Reports module helps users to:

- Access company annual reports from a single interface.
- Review detailed financial disclosures.
- Support financial analysis with official company documents.
- Improve understanding of a company's business performance and strategy.

Features

The Annual Reports module includes:

- Company-wise annual report links
- Easy access to financial documents
- Organized list of available reports
- Direct navigation to supporting resources

How to Use

1. Open the **Annual Reports** module from the navigation menu.
2. Select the company whose report you want to view.
3. Browse the available annual reports or related documents.
4. Open the required report for detailed financial information.
5. Use the report alongside the dashboard's financial metrics for a comprehensive analysis.

Benefits

The Annual Reports module enables users to:

- Access official company disclosures quickly.
- Verify financial information using published reports.
- Gain deeper insights into business operations and management strategies.
- Support investment research with reliable documentation.

Best Practices

- Review annual reports together with the **Company Profile** and **Trend Analysis** modules for a more complete understanding of a company's performance.
- Use annual reports to validate financial figures and gain additional context on significant business events.

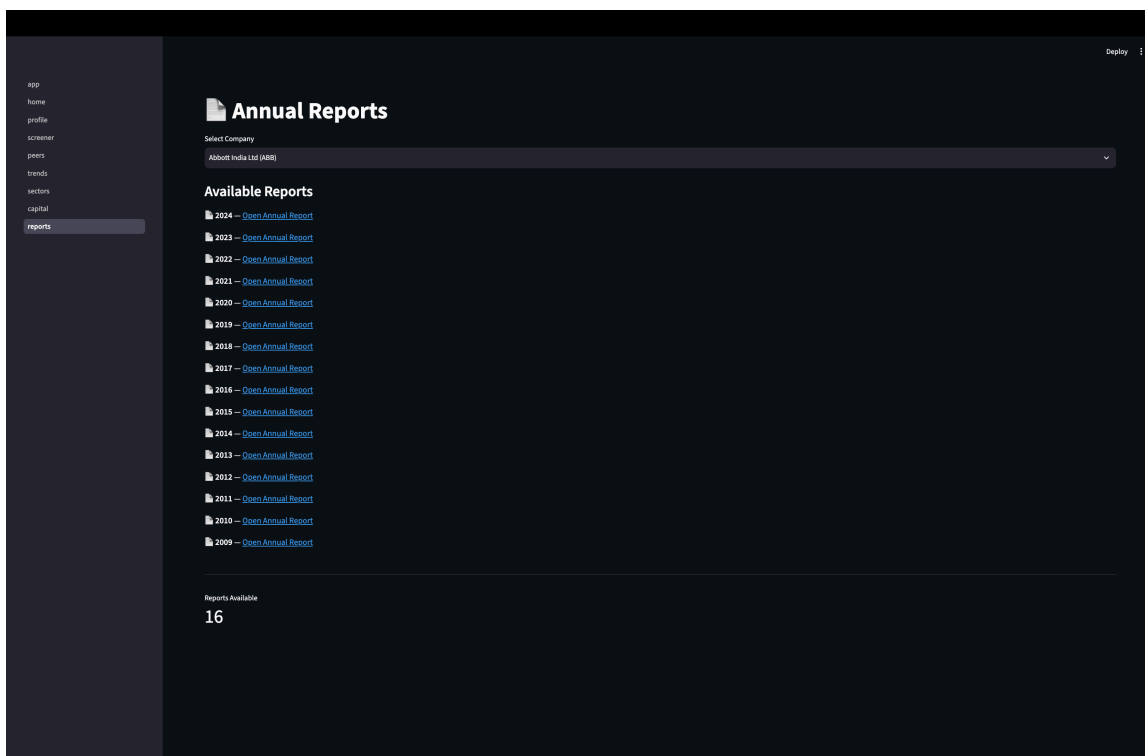


Figure 11: Annual Reports module providing access to company reports and supporting financial documents.

REST API Guide

The **Nifty 100 Analytics Dashboard** provides a REST API built using **FastAPI**. The API allows the dashboard and external applications to access financial data stored in the SQLite database through standardized HTTP requests.

Each endpoint returns data in **JSON** format, making it easy to integrate with web applications, dashboards, or other software systems.

API Base URL

When running the application locally, the API can be accessed at:

`http://127.0.0.1:8000`

Interactive API documentation (Swagger UI):

`http://127.0.0.1:8000/docs`

Common API Endpoints

Endpoint	Method	Description
/	GET	Returns the API welcome message.
/health	GET	Checks whether the API is running correctly.
/companies	GET	Retrieves the list of companies available in the database.
/companies/{company}	GET	Returns the selected company's profile and financial information.
/sectors	GET	Retrieves the list of sectors.
/screener	GET	Returns companies that match the selected screening criteria.
/portfolio	GET	Retrieves portfolio analytics data.
/valuation	GET	Returns valuation-related information.
/peers	GET	Retrieves peer comparison data.
/documents	GET	Returns links to company documents and annual reports.

Example API Request

Example request:

```
GET /health
```

Example response:

```
{  
  "status": "healthy"  
}
```

Benefits of the REST API

The REST API enables:

- Communication between the frontend and backend.
- Access to financial data in JSON format.
- Easy integration with external applications.
- Fast and scalable data retrieval.
- Consistent access to company and financial information.

API Documentation

FastAPI automatically generates interactive API documentation through **Swagger UI**. Users can explore available endpoints, view request parameters, and test API responses directly from the browser without writing additional code.

Nifty100 Analytics API ^{0.1.0} ^{OpenAPI}

REST API for Nifty100 Analytics Platform

Clusters		^
GET	/api/v1/clusters	Get clustered companies
GET	/api/v1/clusters/report	Cluster summary report
GET	/api/v1/clusters/profiles	Cluster feature profiles
GET	/api/v1/clusters/sectors	Cluster sector distribution
Companies		^
GET	/api/v1/companies	Get Companies
GET	/api/v1/companies/{ticker}	Get Company Profile
GET	/api/v1/companies/{ticker}/pl	Get Profit Loss
GET	/api/v1/companies/{ticker}/bs	Get Balance Sheet
GET	/api/v1/companies/{ticker}/cashflow	Get Cashflow
GET	/api/v1/companies/{ticker}/ratios	Get Ratios
GET	/api/v1/companies/{ticker}/tarsheet	Get Tarsheet
Screener		^
GET	/api/v1/screener	Screeners
Sectors		^
GET	/api/v1/sectors	Get Sectors
GET	/api/v1/sectors/{sector}/companies	Get Sector Companies
Peers		^
GET	/api/v1/peers/{group_name}	Get Peer Group
GET	/api/v1/companies/{ticker}/peers/compare	Compare With Peers
Valuation		^
GET	/api/v1/market-cap/{ticker}	Get Market Cap History
Portfolio		^
GET	/api/v1/portfolio/status	Portfolio Status
Documents		^
GET	/api/v1/companies/{ticker}/documents	Get Documents
Health		^
GET	/api/v1/health	Health Check
Default		^

GET	/api/v1/companies/{ticker}/pl	Get Profit Loss
GET	/api/v1/companies/{ticker}/bs	Get Balance Sheet
GET	/api/v1/companies/{ticker}/cashflow	Get Cashflow
GET	/api/v1/companies/{ticker}/ratios	Get Ratios
GET	/api/v1/companies/{ticker}/tarsheet	Get Tarsheet
Screener		^
GET	/api/v1/screener	Screeners
Sectors		^
GET	/api/v1/sectors	Get Sectors
GET	/api/v1/sectors/{sector}/companies	Get Sector Companies
Peers		^
GET	/api/v1/peers/{group_name}	Get Peer Group
GET	/api/v1/companies/{ticker}/peers/compare	Compare With Peers
Valuation		^
GET	/api/v1/market-cap/{ticker}	Get Market Cap History
Portfolio		^
GET	/api/v1/portfolio/status	Portfolio Status
Documents		^
GET	/api/v1/companies/{ticker}/documents	Get Documents
Health		^
GET	/api/v1/health	Health Check
Default		^
GET	/	Root

Schemas		^
Cluster	object	
ClusterListResponse	object	
ClusterProfile	object	
ClusterReport	object	
ClusterSectorDistribution	object	
HTTPValidationError	object	
ValidationError	object	

Figure 12: FastAPI Swagger UI displaying the available REST API endpoints for the Nifty 100 Analytics Dashboard.

Troubleshooting & Best Practices

Troubleshooting

The following table lists common issues that users may encounter while using the Nifty 100 Analytics Dashboard and their recommended solutions.

Issue	Possible Cause	Solution
Dashboard does not load	Streamlit server is not running	Start the Streamlit application and refresh the browser
API request fails	FastAPI server is not running	Start the FastAPI server and verify the API URL
No company data is displayed	Database is missing	Ensure the SQLite database is available and populated with data
Charts are not displayed	Data unavailable or loading issue	Verify that the required financial data exists in the database
Unable to access API documentation	Incorrect API URL or server stopped	Open <code>http://127.0.0.1:8000/docs</code> after starting the FastAPI server.

Best Practices

To achieve the best experience while using the dashboard, follow these recommendations:

- Keep the SQLite database updated with the latest processed data.
- Verify financial information before making investment or research decisions.
- Compare companies within the same sector for meaningful analysis.
- Use multiple dashboard modules together for comprehensive insights.
- Review historical financial trends rather than relying on a single year's performance.
- Refer to annual reports for additional context on company performance.
- Run automated tests before deploying updates to ensure application stability.

System Requirements

The dashboard has been developed and tested using the following technologies:

Component	Technology
Programming Language	Python
Backend Framework	FastAPI
Frontend Framework	Streamlit
Database	SQLite
Data Processing	Pandas
Visualization	Plotly
Testing	Pytest

Conclusion

The **Nifty 100 Analytics Dashboard** provides a centralized platform for analyzing financial data of Nifty 100 companies. By combining interactive dashboards, financial statements, stock screening, peer comparison, sector analysis, capital allocation insights, and REST APIs, the application enables users to perform comprehensive financial analysis efficiently.

This guide has introduced the major features of the dashboard, explained how to navigate its modules, and provided best practices for effective usage. With these tools, users can explore company performance, compare industry peers, analyze trends, and make informed, data-driven decisions.

Acknowledgement

This project was developed as part of the **Bluestock Fintech Internship Capstone Project**.

Developed by:
Mohammad Esa Mohibuddin